

STEP-2

For JEE (Main) & State Engineering Entrance Exams

Assignment

* Questions to be discussed in Class.

Objective Type Questions (One option is correct)

1. A person moves 30 m north, then 20 m east, then $30\sqrt{2}$ m south-west, his displacement is
 (1) 14 m south-west (2) 28 m south
 (3) 10 m west (4) 15 m east
2. The position vector of particle changes from 10 m east to 10 m north in one second. What is the average velocity of the particle?
 (1) 20 ms^{-1} (2) 5 ms^{-1}
 (3) $10\sqrt{2} \text{ ms}^{-1}$ (4) $\left(\frac{10}{\sqrt{2}}\right) \text{ ms}^{-1}$
3. The position co-ordinates of an object are given by $x = 2t^2$, $y = t^2 - 4t$, $z = 3t - 5$. The average velocity of the particle during the interval 0–1 s is
 (1) 10 unit (2) 12 unit
 (3) $\sqrt{22}$ unit (4) 8 unit
4. A projectile has same range for two angles of projection. If times of flight in two cases are t_1 and t_2 then the range of the projectile is
 (1) $\frac{1}{2}gt_1t_2$ (2) $\frac{1}{4}gt_1t_2$
 (3) gt_1t_2 (4) $\frac{1}{8}gt_1t_2$
5. A projectile can have the same range for two angle of projection. If times of flight in two cases are t_1 and t_2 then velocity of projection of the projectile is
 (1) $\frac{1}{2}g(t_1 + t_2)$ (2) $\frac{1}{4}g(t_1 + t_2)$
 (3) $\frac{1}{2}g(t_1^2 + t_2^2)^{\frac{1}{2}}$ (4) $\frac{1}{2}g(t_1^2 + t_2^2)$
6. A particle is thrown with the speed u at an angle α with the horizontal. When the particle makes an angle β with the horizontal, its speed will be
 (1) $u \cos \alpha$ (2) $u \cos \alpha \sec \beta$
 (3) $u \cos \alpha \cos \beta$ (4) $u \sec \alpha \cos \beta$
7. A particle is projected with velocity 50 m/s at an angle 60° with the horizontal from the ground. The time after which its velocity will make an angle 45° with the horizontal is
 (1) 2.5 s (2) 1.83 s
 (3) 2.37 s (4) 3.72 s
8. A projectile projected from the ground has its direction of motion making an angle $\frac{\pi}{4}$ with the horizontal at a height 40 m. Its initial velocity of projection is 50 m/s, the angle of projection is
 (1) $\frac{1}{2} \cos^{-1}\left(-\frac{8}{25}\right)$
 (2) $\frac{1}{2} \cos^{-1}\left(+\frac{8}{25}\right)$
 (3) $\frac{1}{2} \cos^{-1}\left(-\frac{4}{5}\right)$
 (4) $\frac{1}{2} \cos^{-1}\left(-\frac{1}{4}\right)$
9. A particle projected at some angle with velocity 50 m/s crosses a 20 m high wall after 4 s from the time of projection. The angle of projection of the particle is
 (1) 30° (2) 45°
 (3) 60° (4) 53°
10. An aeroplane is moving horizontally with a velocity of u . It drops a packet from a height h . The time taken the packet in reaching the ground will be
 (1) $\sqrt{\left(\frac{2g}{h}\right)}$ (2) $\sqrt{\left(\frac{2u}{g}\right)}$
 (3) $\sqrt{\left(\frac{h}{2g}\right)}$ (4) $\sqrt{\left(\frac{2h}{g}\right)}$

11. A body of mass m thrown horizontally with a velocity v from the top of the tower of height h touches the ground at a distance 250 m from the foot of tower. Another body of mass $2m$ thrown horizontally with half the velocity from a tower of height $4h$ will touch the ground at a distance

- (1) 250 m (2) 1000 m
(3) 125 m (4) 500 m

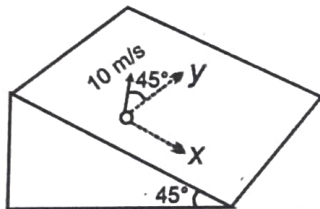
12. A stair case contains three steps having height 10 cm each. If each are 20 cm wide then the minimum horizontal velocity of a ball moving off the upper mark plane so as to hit the lowest plane directly is

- (1) 4 m/s (2) 2 m/s
(3) 1 m/s (4) 0.5 m/s

13. From the top of a tower of height 40 m a ball is projected upward with a speed of 20 ms^{-1} at an elevation of 30° . The ratio of the total time taken by the ball to hit the ground to its time to come back at the same level is ($g = 10 \text{ ms}^{-2}$)

- (1) 2 : 1 (2) 3 : 1
(3) 3 : 2 (4) 4 : 1

- 14*. A small marble is projected with a velocity of 10 m/s in a direction 45° from the horizontal y -direction, along the smooth wedge. Calculate the magnitude v of its velocity after 2 seconds. Take $g = 10 \text{ m/s}^2$. (Note that the particle is moving in inclined plane of the wedge)



- (1) 20 m/s (2) 15 m/s
(3) $10\sqrt{3}$ m/s (4) 10 m/s

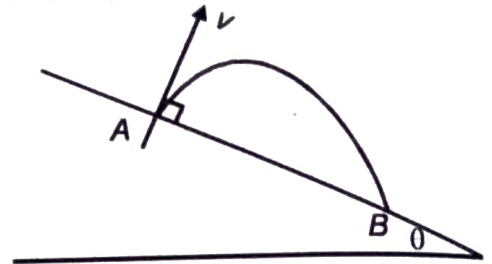
15. A projectile is fired horizontally from an inclined plane (of inclination 45° with horizontal) with speed 50 m/s. If $g = 10 \text{ m/s}^2$, the range measured along the incline is

- (1) 500 m (2) $500\sqrt{2}$ m
(3) $200\sqrt{2}$ m (4) $500/\sqrt{2}$ m

16. At what angle with an incline plane of inclination 30° a particle should be projected to have maximum range on the inclined plane?

- (1) 30° (2) 45°
(3) 15° (4) 75°

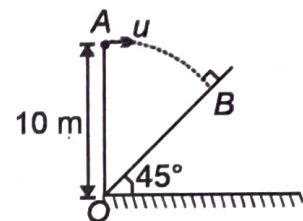
17. If time taken by the projectile to reach B is T , then AB is equal to



- (1) $Tv \sin \theta$
(2) $Tv \cos \theta$
(3) $Tv \sec \theta$
(4) $Tv \tan \theta$

Paragraph for Q. Nos. 18 to 20*

A particle is projected horizontally with speed u from point A, which is 10 m above the ground. If the particle hits the inclined plane perpendicularly at point B. [$g = 10 \text{ m/s}^2$]



- 18*. Find horizontal speed with which the particle was projected.

- (1) $\frac{20}{3}$ (2) $20\sqrt{\frac{20}{3}}$
(3) $u = 10\sqrt{\frac{2}{3}}$ (4) $10\sqrt{\frac{3}{2}}$

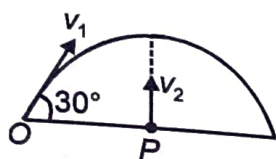
- 19*. Find the length OB along the inclined plane.

- (1) $\frac{10}{\sqrt{3}}$ (2) $20\sqrt{3}$
(3) $\frac{20}{\sqrt{3}}$ (4) $\frac{20\sqrt{2}}{3}$

- 20*. Speed at point B will be

- (1) $\frac{20}{\sqrt{3}}$
(2) $20\sqrt{3}$
(3) $\frac{10}{\sqrt{3}}$
(4) $10\sqrt{3}$

21. A particle is projected from point O with velocity v_1 at an angle 30° with the horizontal and another particle is projected from point P which is vertically below the highest point with velocity v_2 in vertically upward direction as shown. For the two particles to collide the ratio $\frac{v_1}{v_2}$ will be



- (1) 1
(2) 2
(3) 0.5
(4) 1.5

Paragraph for Q. Nos. 22 to 24

When a boat travels in a river (strictly in a straight line), it can go either in the direction of flow of river (i.e., downstream) or in the direction opposite the flow of river (i.e., upstream). Thus the boat's actual speed is more than by which it can move in stationary water while travelling downstream (as river's flow speed is added to it) and less while travelling upstream (as the boat moves against the flow of river). Based on the given information answer the following questions.

A boat going downstream in a flowing river overcame a raft at a point P. 1 h later it turned back and after some time passed the raft at a distance 6 km from point P.

22. After reversing its direction, how much time was taken by the boat to meet the raft again (i.e., 2nd time)?

- (1) 2 h
(2) 1 h
(3) 30 min
(4) 3 h

23. Find the speed of river.

- (1) 4 km/h
(2) 2 km/h
(3) 3 km/h
(4) 1 km/h

24. Now, if instead of 6 km they have met at 8 km from point P, find the speed of river.

- (1) 1 km/h
(2) 2 km/h
(3) 3 km/h
(4) 4 km/h

Paragraph for Q. Nos. 25 to 27

Raindrops are falling with velocity $10\sqrt{2}$ m/s making an angle 45° with the vertical. The drops appear to be falling vertically to a man running with constant velocity. The velocity of rain drops change such that the rain drops now appear to be falling vertically with $\sqrt{3}$ times the velocity it appeared earlier to the same person running with same velocity.

25. The magnitude of velocity of man w.r.t. ground is

- (1) $10\sqrt{2}$ m/s
(2) $10\sqrt{3}$ m/s
(3) 10 m/s
(4) 20 m/s

26. After the velocity of rain drops change the magnitude of velocity of raindrops with respect to ground is

- (1) $20\sqrt{3}$ m/s
(2) 10 m/s
(3) $10\sqrt{3}$ m/s
(4) 20 m/s

27. The angle (in degrees) between the initial and final velocity of rain drops with respect to ground

- (1) 15°
(2) 8°
(3) 22.5°
(4) 37°

- 28*. What is the maximum magnitude of change in velocity of a motorcycle moving with a uniform speed v_0 in a circular path of length $l = \frac{\pi}{3}R$ and radius R ? Treat motorcycle as a particle.

- (1) $|\Delta \vec{v}| = \frac{v_0}{2}$
(2) $|\Delta \vec{v}| = \sqrt{3}v_0$
(3) $|\Delta \vec{v}| = 2v_0$
(4) $|\Delta \vec{v}| = v_0$

29. A train 1 moves from east to west (clockwise) and another train 2 moves from west to east (anticlockwise) on the equator with equal speeds relative to ground. The ratio of their centripetal acceleration $\frac{a_1}{a_2}$ relative to centre of earth is

- (1) > 1
(2) $= 1$
(3) < 1
(4) ≥ 1

30. A particle is performing circular motion in a circle of radius R in such a way that at every moment magnitude of its tangential and radial acceleration is equal. If initial velocity of particle is v_0 then time period of 1st revolution is

- (1) $\frac{R}{v_0}$
(2) $\frac{R}{v_0}(1 - e^{-2\pi})$
(3) $\frac{R}{v_0}(e^{-2\pi})$
(4) $\frac{2\pi R}{v_0}$

Numerical Value Based Questions

31. A particle moves along a curve $y = x^2$ with constant speed of 10 m/s. When the particle is at origin, then the radius of curvature of path is n m. The value of n is _____.

32*. Rain is falling vertically with a speed of 4 m/s. After some time, wind starts blowing with a speed of 3 m/s in the north to south direction. In order to protect himself from rain, a man standing on the ground should hold his umbrella at an angle θ given by $\theta = \tan^{-1}\left(\frac{m}{n}\right)$ with vertical towards north. If m, n are coprime then value of mn is _____.

33. A gun fitted on the top of a moving car is aimed in the backward direction at an angle 60° with the horizontal. If the muzzle velocity of the bullet fired from the gun is 8 m/s, then the speed of the car for which the bullet comes out vertically is n m/s. The value of n is _____.

34. A particle is performing motion whose position vector is given by $\vec{r} = 15 (\cos pt \hat{i} + \sin pt \hat{j})$ m, when p is in rad/s and t is in seconds. The magnitude of centripetal acceleration at $t = 3$ s, is np^2 m/s². The value of n is _____.

35. Speed of a particle moving on a circular path of radius 2 m is varying with time as $v = (2t^2)$ m/s. Net acceleration of the particle at $t = 2$ second is $8\sqrt{n}$ m/s². The value of n is _____.

PREVIOUS YEARS QUESTIONS

Objective Type Questions

[Motion in Plane]

36. A particle is moving with a velocity $\vec{v} = K(y\hat{i} + x\hat{j})$, where K is a constant. The general equation for its path is [JEE (Main)-2019]

- (1) $y^2 = x + \text{constant}$ (2) $y = x^2 + \text{constant}$
(3) $y^2 = x^2 + \text{constant}$ (4) $xy = \text{constant}$

37. The position co-ordinates of a particle moving in a 3-D coordinate system is given by $x = a \cos \omega t$, $y = a \sin \omega t$ and $z = a\omega t$. The speed of the particle is [JEE (Main)-2019]

- (1) $2a\omega$ (2) $\sqrt{2}a\omega$
(3) $\sqrt{3}a\omega$ (4) $a\omega$

38. A person standing on an open ground hears the sound of a jet aeroplane, coming from north at an angle 60° with ground level. But he finds the aeroplane right vertically above his position. If v is the speed of sound, speed of the plane is [JEE (Main)-2019]

(1) $\frac{2v}{\sqrt{3}}$

(2) $\frac{\sqrt{3}}{2}v$

(3) $\frac{v}{2}$

(4) v

39. The position vector of a particle changes with time according to the relation $\vec{r}(t) = 15t^2\hat{i} + (4 - 20t^2)\hat{j}$. What is the magnitude of the acceleration at $t = 1$? [JEE (Main)-2019]

(1) 50

(2) 100

(3) 40

(4) 25

40. A particle of mass m is moving along a trajectory given by

$x = x_0 + a \cos \omega_1 t$

$y = y_0 + b \sin \omega_2 t$

The torque, acting on the particle about the origin, at $t = 0$ is : [JEE (Main)-2019]

(1) $-m(x_0 b \omega_2^2 - y_0 a \omega_1^2) \hat{k}$

(2) $m(-x_0 b + y_0 a) \omega_1^2 \hat{k}$

(3) $+m y_0 a \omega_1^2 \hat{k}$

(4) Zero

41. A particle starts from the origin at $t = 0$ with an initial velocity of $3.0\hat{i}$ m/s and moves in the x - y plane with a constant acceleration $(6.0\hat{i} + 4.0\hat{j})$ m/s². The x -coordinate of the particle at the instant when its y -coordinate is 32 m is D meters. The value of D is [JEE (Main)-2020]

(1) 60

(2) 32

(3) 40

(4) 50

42. Starting from the origin at time $t = 0$, with initial velocity $5\hat{j}$ ms⁻¹, a particle moves in the x - y plane with a constant acceleration of $(10\hat{i} + 4\hat{j})$ ms⁻². At time t , its coordinates are $(20 \text{ m}, y_0 \text{ m})$. The values of t and y_0 are, respectively [JEE (Main)-2020]

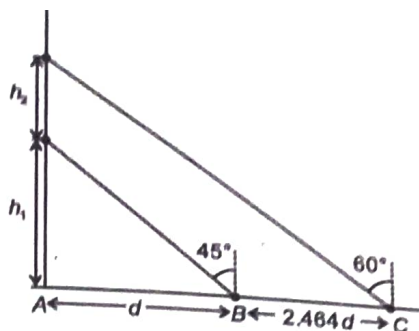
(1) 4 s and 52 m

(2) 2 s and 24 m

(3) 5 s and 25 m

(4) 2 s and 18 m

43*. A balloon is moving up in air vertically above a point A on the ground. When it is at a height h_1 , a girl standing at a distance d (point B) from A (see figure) sees it at an angle 45° with respect to the vertical. When the balloon climbs up a further height h_2 , it is seen at an angle 60° with respect to the vertical if the girl moves further by a distance $2.464d$ (point C). Then the height h_2 is (given $\tan 30^\circ = 0.5774$) [JEE (Main)-2020]



- (1) d (2) $0.732d$
(3) $1.464d$ (4) $0.464d$

44. A mosquito is moving with a velocity $\vec{v} = 0.5t^2\hat{i} + 3t\hat{j} + 9\hat{k}$ m/s and accelerating in uniform conditions. What will be the direction of motion of mosquito after 2 s?

[JEE (Main)-2021]

- (1) $\tan^{-1}\left(\frac{2}{3}\right)$ from y-axis
(2) $\cos^{-1}\left(\frac{6}{11}\right)$ from y-axis
(3) $\tan^{-1}\left(\frac{5}{2}\right)$ from x-axis
(4) $\tan^{-1}\left(\frac{2}{3}\right)$ from x-axis

45. Motion of a particle in x-y plane is described by a set of following equations $x = 4\sin\left(\frac{\pi}{2} - \omega t\right)$ m and $y = 4\sin(\omega t)$ m. The path of the particle will be

[JEE (Main)-2022]

- (1) Circular (2) Helical
(3) Parabolic (4) Elliptical

46. At time $t = 0$ a particle starts travelling from a height $7\hat{z}$ cm in a plane keeping z coordinate constant. At any instant of time its position along the $\hat{x}$ and $\hat{y}$ directions are defined at $3t$ and $5t^3$ respectively. At $t = 1$ s acceleration of the particle will be

[JEE (Main)-2022]

- (1) $-30\hat{y}$ (2) $30\hat{y}$
(3) $3\hat{x} + 15\hat{y}$ (4) $3\hat{x} + 15\hat{y} + 7\hat{z}$

[Projectile Motion]

47. Two guns A and B can fire bullets at speeds 1 km/s and 2 km/s respectively. From a point on a horizontal ground, they are fired in all possible directions. The ratio of maximum areas covered by the bullets fired by the two guns, on the ground is

[JEE (Main)-2019]

- (1) 1 : 4 (2) 1 : 8
(3) 1 : 2 (4) 1 : 16

48. A shell is fired from a fixed artillery gun with an initial speed u such that it hits the target on the ground at a distance R from it. If t_1 and t_2 are the values of the time taken by it to hit the target in two possible ways, the product $t_1 t_2$ is

[JEE (Main)-2019]

- (1) $\frac{R}{2g}$ (2) $\frac{2R}{g}$
(3) $\frac{R}{g}$ (4) $\frac{R}{4g}$

49. Two particles are projected from the same point with the same speed u such that they have the same range R , but different maximum heights, h_1 and h_2 . Which of the following is correct?

[JEE (Main)-2019]

- (1) $R^2 = 4h_1 h_2$ (2) $R^2 = 16h_1 h_2$
(3) $R^2 = 2h_1 h_2$ (4) $R^2 = h_1 h_2$

50. The trajectory of a projectile in a vertical plane is $y = \alpha x - \beta x^2$, where α and β are constants and x & y are respectively the horizontal and vertical distances of the projectile from the point of projection. The angle of projection θ and the maximum height attained H are respectively given by

[JEE (Main)-2021]

- (1) $\tan^{-1}\alpha, \frac{4\alpha^2}{\beta}$ (2) $\tan^{-1}\beta, \frac{\alpha^2}{2\beta}$
(3) $\tan^{-1}\left(\frac{\beta}{\alpha}\right), \frac{\alpha^2}{\beta}$ (4) $\tan^{-1}\alpha, \frac{\alpha^2}{4\beta}$

51. A player kicks a football with an initial speed of 25 ms^{-1} at an angle of 45° from the ground. What are the maximum height and the time taken by the football to reach at the highest point during motion? (Take $g = 10 \text{ ms}^{-2}$)

[JEE (Main)-2021]

- (1) $h_{\max} = 10 \text{ m}, T = 2.5 \text{ s}$
(2) $h_{\max} = 15.625 \text{ m}, T = 1.77 \text{ s}$
(3) $h_{\max} = 3.54 \text{ m}, T = 0.125 \text{ s}$
(4) $h_{\max} = 15.625 \text{ m}, T = 3.54 \text{ s}$

52. A projectile is projected with velocity of 25 m/s at an angle θ with the horizontal. After t seconds its inclination with horizontal becomes zero. If R represents horizontal range of the projectile, the value of θ will be [use $g = 10 \text{ m/s}^2$]

[JEE (Main)-2022]

- (1) $\frac{1}{2} \sin^{-1}\left[\frac{5t^2}{4R}\right]$ (2) $\frac{1}{2} \sin^{-1}\left[\frac{4R}{5t^2}\right]$
(3) $\tan^{-1}\left[\frac{4t^2}{5R}\right]$ (4) $\cot^{-1}\left[\frac{R}{20t^2}\right]$

53. A projectile is launched at an angle ' α ' with the horizontal with a velocity 20 ms^{-1} . After 10 s, its inclination with horizontal is ' β '. The value of $\tan\beta$ will be ($g = 10 \text{ ms}^{-2}$).

[JEE (Main)-2022]

- (1) $\tan\alpha + 5\sec\alpha$ (2) $\tan\alpha - 5\sec\alpha$
(3) $2\tan\alpha - 5\sec\alpha$ (4) $2\tan\alpha + 5\sec\alpha$

54. A ball is projected from the ground with a speed 15 ms^{-1} at an angle θ with horizontal so that its range and maximum height are equal, then ' $\tan\theta$ ' will be equal to

[JEE (Main)-2022]

- (1) $\frac{1}{4}$ (2) $\frac{1}{2}$
(3) 2 (4) 4

55. A ball is projected with kinetic energy E , at an angle of 60° to the horizontal. The kinetic energy of this ball at the highest point of its flight will become

[JEE (Main)-2022]

- (1) Zero (2) $\frac{E}{2}$
(3) $\frac{E}{4}$ (4) E

56. Two projectiles are thrown with same initial velocity making an angle of 45° and 30° with the horizontal respectively. The ratio of their respective ranges will be

[JEE (Main)-2022]

- (1) $1:\sqrt{2}$ (2) $\sqrt{2}:1$
(3) $2:\sqrt{3}$ (4) $\sqrt{3}:2$

57. The maximum vertical height to which a man can throw a ball is 136 m. The maximum horizontal distance upto which he can throw the same ball is:

[JEE (Main)-2023]

- (1) 272 m (2) 68 m
(3) 192 m (4) 136 m

58. Two objects are projected with same velocity ' u ' however at different angles α and β with the horizontal. If $\alpha + \beta = 90^\circ$, the ratio of horizontal range of the first object to the 2nd object will be

[JEE (Main)-2023]

- (1) 1 : 2 (2) 4 : 1
(3) 2 : 1 (4) 1 : 1

59. A stone is projected at angle 30° to the horizontal. The ratio of kinetic energy of the stone at point of projection to its kinetic energy at the highest point of flight will be

[JEE (Main)-2023]

- (1) 4 : 3

- (2) 4 : 1

- (3) 1 : 2

- (4) 1 : 4

60. Projectiles A and B are thrown at angles of 45° and 60° with vertical respectively from top of a 400 m high tower. If their ranges and times of flight are same, the ratio of their speeds of projection $v_A : v_B$ is :

[JEE (Main)-2024]

[Take $g = 10 \text{ m/s}^2$]

- (1) $1:\sqrt{3}$ (2) $1:\sqrt{2}$
(3) $\sqrt{2}:1$ (4) 1 : 2

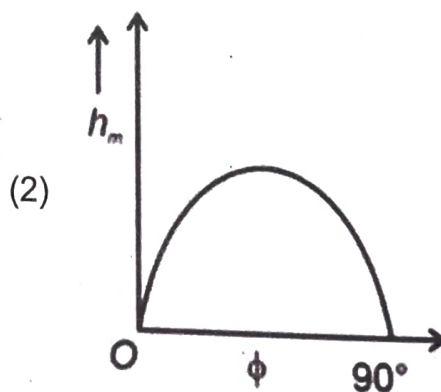
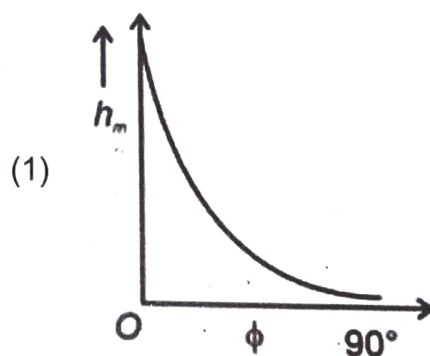
61. Two projectiles are fired from ground with same initial speeds from same point at angles $(45^\circ + \alpha)$ and $(45^\circ - \alpha)$ with horizontal direction. The ratio of their times of flights is

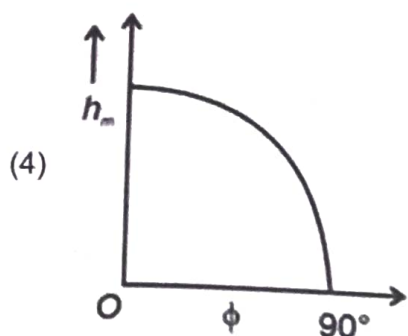
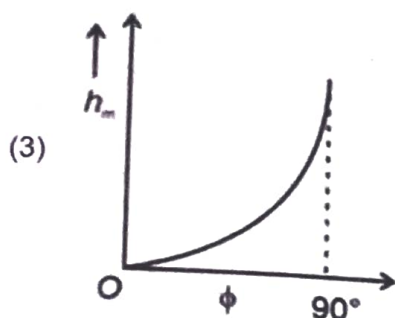
[JEE (Main)-2025]

- (1) $\frac{1 + \sin 2\alpha}{1 - \sin 2\alpha}$ (2) $\frac{1 + \tan \alpha}{1 - \tan \alpha}$
(3) $\frac{1 - \tan \alpha}{1 + \tan \alpha}$ (4) 1

62. The angle of projection of a particle is measured from the vertical axis as ϕ and the maximum height reached by the particle is h_m . Here h_m as function of ϕ can be presented as

[JEE (Main)-2025]





[Circular Motion]

63. A clock has a continuously moving second's hand of 0.1 m length. The acceleration of the tip of the hand (in units of ms^{-2}) is of the order of

[JEE (Main)-2020]

- (1) 10^{-1} (2) 10^{-2}
(3) 10^{-4} (4) 10^{-3}

64. A ball is spun with angular acceleration $\alpha = 6t^2 - 2t$, where t is in second and α is in rads^{-2} . At $t = 0$, the ball has angular velocity of 10 rads^{-1} and angular position of 4 rad. The most appropriate expression for the angular position of the ball is :

[JEE (Main)-2022]

- (1) $\frac{3}{4}t^4 - t^2 + 10t$ (2) $\frac{t^4}{2} - \frac{t^3}{3} + 10t + 4$
(3) $\frac{2t^4}{3} - \frac{t^3}{6} + 10t + 12$ (4) $2t^4 - \frac{t^3}{2} + 5t + 4$

65. An object moves at a constant speed along a circular path in a horizontal plane with centre at the origin. When the object is at $x = +2 \text{ m}$, its velocity is $-4\hat{j} \text{ m/s}$. The object's velocity (v) and acceleration (a) at $x = -2 \text{ m}$ will be

[JEE (Main)-2023]

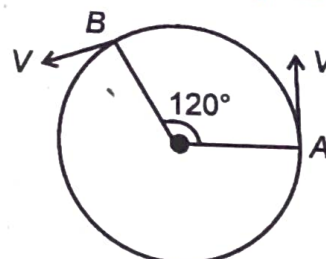
- (1) $v = 4\hat{i} \text{ m/s}$, $a = 8\hat{j} \text{ m/s}^2$
(2) $v = 4\hat{j} \text{ m/s}$, $a = 8\hat{i} \text{ m/s}^2$

(3) $v = -4\hat{i} \text{ m/s}$, $a = -8\hat{j} \text{ m/s}^2$

(4) $v = -4\hat{j} \text{ m/s}$, $a = 8\hat{i} \text{ m/s}^2$

- 66*. As shown in the figure, a particle is moving with constant speed $\pi \text{ m/s}$. Considering its motion from A to B, the magnitude of the average velocity is:

[JEE (Main)-2023]



- (1) $\sqrt{3} \text{ m/s}$ (2) $\pi \text{ m/s}$
(3) $1.5\sqrt{3} \text{ m/s}$ (4) $2\sqrt{3} \text{ m/s}$

67. A particle moving in a circle of radius R with uniform speed takes time T to complete one revolution. If this particle is projected with the same speed at an angle θ to the horizontal, the maximum height attained by it is equal to $4R$. The angle of projection θ is then given by:

[JEE (Main)-2024]

- (1) $\cos^{-1}\left[\frac{2gT^2}{\pi^2 R}\right]^{\frac{1}{2}}$ (2) $\sin^{-1}\left[\frac{2gT^2}{\pi^2 R}\right]^{\frac{1}{2}}$
(3) $\sin^{-1}\left[\frac{\pi^2 R}{2gT^2}\right]^{\frac{1}{2}}$ (4) $\cos^{-1}\left[\frac{\pi R}{2gT^2}\right]^{\frac{1}{2}}$

[Relative Velocity]

68. The stream of a river is flowing with a speed of 2 km/h. A swimmer can swim at a speed of 4 km/h. What should be the direction of the swimmer with respect to the flow of the river to cross the river straight?

[JEE (Main)-2019]

- (1) 60° (2) 90°
(3) 150° (4) 120°

69. When a car is at rest, its driver sees rain drops falling on it vertically. When driving the car with speed v , he sees that rain drops are coming at an angle 60° from the horizontal. On further increasing the speed of the car to $(1 + \beta)v$, this angle changes to 45° . The value of β is close to

[JEE (Main)-2020]

- (1) 0.37 (2) 0.41
(3) 0.73 (4) 0.50

70. A girl standing on road holds her umbrella at 45° with the vertical to keep the rain away. If she starts running without umbrella with a speed of $15\sqrt{2} \text{ kmh}^{-1}$, the rain drops hit her head vertically. The speed of rain drops with respect to the moving girl is [JEE (Main)-2022]

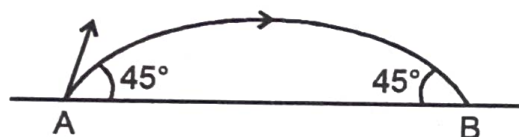
- (1) 30 kmh^{-1} (2) $\frac{25}{\sqrt{2}} \text{ kmh}^{-1}$
 (3) $\frac{30}{\sqrt{2}} \text{ kmh}^{-1}$ (4) 25 kmh^{-1}

Numerical Value Based Questions

71. A particle is moving along the x -axis with its coordinate with time ' t ' given by $x(t) = 10 + 8t - 3t^2$. Another particle is moving along the y -axis with its coordinate as a function of time given by $y(t) = 5 - 8t^3$. At $t = 1 \text{ s}$, the speed of the second particle as measured in the frame of the first particle is given as $\sqrt{v}$. Then v (in m/s) is _____.

[JEE (Main)-2020]

- 72*. The projectile motion of a particle of mass 5 g is shown in the figure.



The initial velocity of the particle is $5\sqrt{2} \text{ ms}^{-1}$ and the air resistance is assumed to be negligible. The magnitude of the change in momentum between the points A and B is $x \times 10^{-2} \text{ kgms}^{-1}$.

The value of x , to the nearest integer, is _____.

[JEE (Main)-2021]

73. A body rotating with an angular speed of 600 rpm is uniformly accelerated to 1800 rpm in 10 sec . The number of rotations made in the process is _____.

[JEE (Main)-2021]

74. A car is moving on a circular path of radius 600 m such that the magnitudes of the tangential acceleration and centripetal acceleration are equal. The time taken by the car to complete first quarter of revolution, if it is moving with an initial speed of 54 km/hr is $t(1 - e^{-\pi/2}) \text{ s}$. The value of t is _____.

[JEE (Main)-2023]

75. The speed of a swimmer is 4 km h^{-1} in still water. If the swimmer makes his strokes normal to the flow of river of width 1 km , he reaches a point 750 m down the stream on the opposite bank.

The speed of the river water is _____ km h^{-1}

[JEE (Main)-2023]

- 76*. Two bodies are projected from ground with same speeds 40 ms^{-1} at two different angles with respect to horizontal. The bodies were found to have same range. If one of the body was projected at an angle of 60° , with horizontal then sum of the maximum heights, attained by the two projectiles, is _____ m . (Given $g = 10 \text{ ms}^{-2}$)

[JEE (Main)-2023]

77. A body of mass M thrown horizontally with velocity v from the top of the tower of height H touches the ground at a distance of 100 m from the foot of the tower. A body of mass $2M$ thrown at a velocity $\frac{v}{2}$ from the top of the tower of height $4H$ will touch the ground at a distance of _____ m .

[JEE (Main)-2024]

78. The maximum speed of a boat in still water is 27 km/h . Now this boat is moving downstream in a river flowing at 9 km/h . A man in the boat throws a ball vertically upwards with speed of 10 m/s . Range of the ball as observed by an observer at rest on the river bank, is _____ cm . (Take $g = 10 \text{ m/s}^2$)

[JEE (Main)-2025]